Step 6 - beta diversity statistics

A: PERMANOVA (adonis, Bray-Curtis, 999 perm.) - our result vs paper

	Our R2	Our F	Our P	Paper R2	Paper F	Paper P
health_status	21.5%	16.33	0.001	17.2%	9.35	0.001
cultivar	22.6%	3.43	0.002	n.s.	-	0.223
health_status:cultivar	8.3%	1.26	0.263	12.4%	1.43	0.016
Residuals	47.5%					
Total	100.0%					

B: PERMANOVA, canker bacteria only (cultivar)

	Df	SumsOfSqs	MeanSqs	F.Model	R2	Pr(>F)
cultivar	5	1.3154	0.2631	1.0	0.1613	0.471
Residuals	26	6.84	0.2631	nan	0.8387	nan
Total	31	8.1553	nan	nan	1.0	nan

Step 6 - alpha diversity tests (one-way ANOVA as in paper; Kruskal-Wallis)

A: canker vs healthy

	ANOVA_F	ANOVA_P	KruskalWallis_H	KW_P
Simpson	47.3	1.4e-08	31.35	2.2e-08
Shannon	32.25	8.8e-07	31.35	2.2e-08
Observed	25.26	8.1e-06	31.43	2.1e-08
Chao1	25.3	8e-06	31.43	2.1e-08
Faith PD	75.59	2.9e-11	31.1	2.4e-08

A: among cultivars - canker

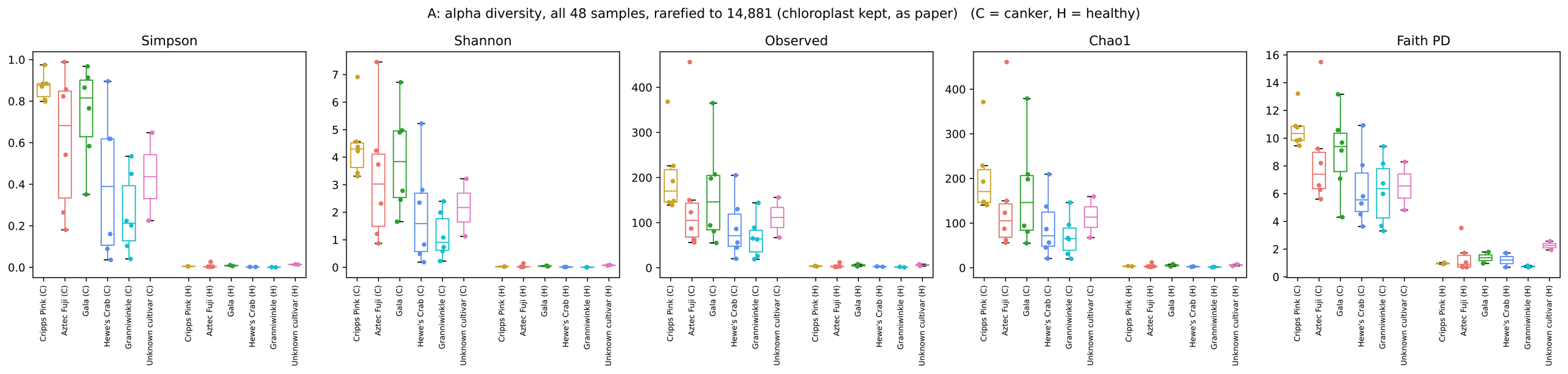
	ANOVA_F	ANOVA_P	KruskalWallis_H	KW_P
Simpson	4.45	0.0046	13.76	0.017
Shannon	2.97	0.03	12.99	0.023
Observed	1.55	0.21	9.73	0.083
Chao1	1.46	0.24	8.84	0.12
Faith PD	2.42	0.063	11.64	0.04

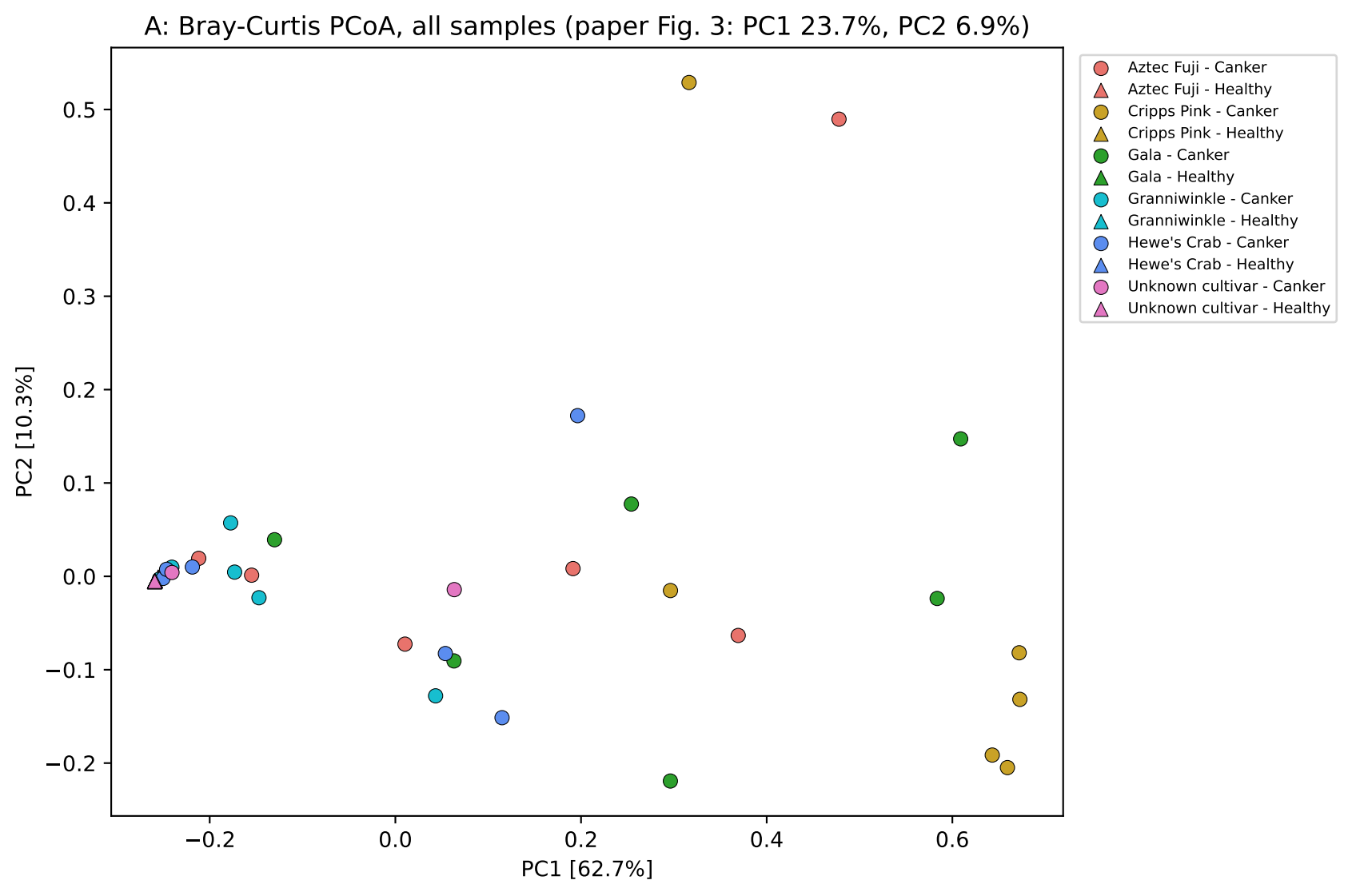
A: among cultivars - healthy

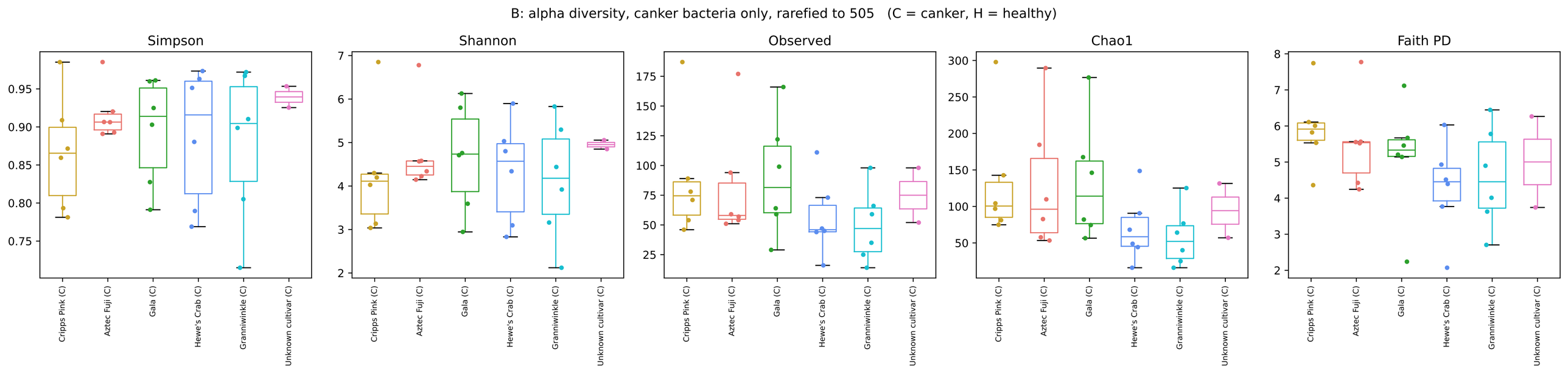
	ANOVA_F	ANOVA_P	KruskalWallis_H	KW_P
Simpson	1.02	0.45	11.47	0.043
Shannon	0.98	0.47	11.47	0.043
Observed	0.62	0.69	7.95	0.16
Chao1	0.62	0.69	7.95	0.16
Faith PD	0.75	0.61	6.13	0.29

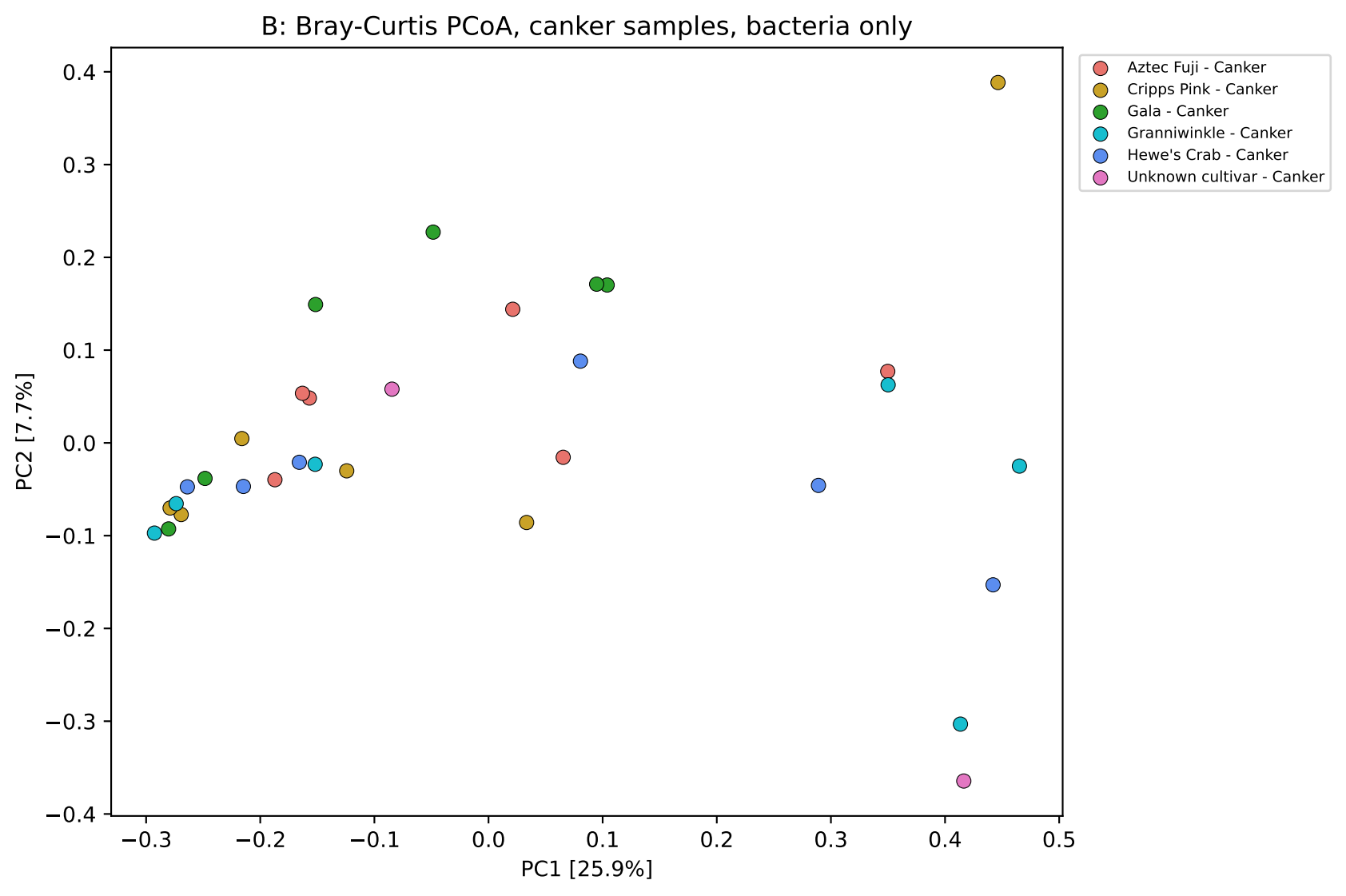
B: among cultivars - canker, bacteria only

	ANOVA_F	ANOVA_P	KruskalWallis_H	KW_P
Simpson	0.46	0.8	2.12	0.83
Shannon	0.31	0.9	2.66	0.75
Observed	0.91	0.49	4.97	0.42
Chao1	1.35	0.28	8.01	0.16
Faith PD	1.16	0.35	5.16	0.4


A: alpha diversity, all 48 samples, rarefied to 14,881 (chloroplast kept, as paper) (C = canker, H = healthy)
Simpson
Shannon
Observed
Chao1
Faith PD
Cripps Pink (C)
Aztec Fuji (C)
Gala (C)
Hewe's Crab (C)
Granniwinkle (C)
Unknown cultivar (C)
Cripps Pink (H)
Aztec Fuji (H)
Gala (H)
Hewe's Crab (H)
Granniwinkle (H)
Unknown cultivar (H)


A: Bray-Curtis PCoA, all samples (paper Fig. 3: PC1 23.7%, PC2 6.9%)
PC1 [62.7%]
PC2 [10.3%]
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.2
0.0
0.2
0.4
0.6
Aztec Fuji - Canker
Aztec Fuji - Healthy
Cripps Pink - Canker
Cripps Pink - Healthy
Gala - Canker
Gala - Healthy
Granniwinkle - Canker
Granniwinkle - Healthy
Hewe's Crab - Canker
Hewe's Crab - Healthy
Unknown cultivar - Canker
Unknown cultivar - Healthy


B: alpha diversity, canker bacteria only, rarefied to 505 (C = canker, H = healthy)
Simpson
Shannon
Observed
Chao1
Faith PD
Cripps Pink (C)
Aztec Fuji (C)
Gala (C)
Hewe's Crab (C)
Granniwinkle (C)
Unknown cultivar (C)


B: Bray-Curtis PCoA, canker samples, bacteria only
Aztec Fuji - Canker
Cripps Pink - Canker
Gala - Canker
Granniwinkle - Canker
Hewe's Crab - Canker
Unknown cultivar - Canker
PC1 [25.9%]
PC2 [7.7%]
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5

Step 6 - alpha diversity summary

A: alpha diversity range by health status (paper: canker Observed 16-470; healthy Shannon 0-2, Simpson 0-0.75)

	Canker	Healthy
('simpson', 'min')	0.04	0.0
('simpson', 'mean')	0.57	0.01
('simpson', 'max')	0.99	0.03
('shannon', 'min')	0.19	-0.0
('shannon', 'mean')	2.92	0.03
('shannon', 'max')	7.46	0.14
('observed_features', 'min')	19.0	1.0
('observed_features', 'mean')	135.09	3.94
('observed_features', 'max')	456.0	12.0
('chao1', 'min')	20.0	1.0
('chao1', 'mean')	137.2	3.94
('chao1', 'max')	460.86	12.0
('faith_pd', 'min')	3.31	0.69
('faith_pd', 'mean')	8.06	1.35
('faith_pd', 'max')	15.49	3.53